



K L Deemed to be University
Department of Computer Science and Information Technology -- KLVZA
Course Handout
2025-2026, Odd Sem

Course Title	:COMPILER DESIGN
Course Code	:23CS2235F
L-T-P-S Structure	: 2-0-2-0
Pre-requisite	:
Credits	: 3
Course Coordinator	:KAVITHA SARIHADDU
Team of Instructors	:
Teaching Associates	:

Syllabus :INTRODUCTION TO COMPILERS: Definition of compiler, interpreter and its differences, the phases of a compiler, role of lexical analyzer, regular expressions, finite automata, from regular expressions to finite automata, pass and phases of translation, bootstrapping, LEX-lexical analyzer generator.

PARSING: Parsing, role of parser, context free grammar, derivations, parse trees, ambiguity, elimination of left recursion, left factoring, eliminating ambiguity from dangling-else grammar, classes of parsing, top down parsing - backtracking, recursive descent parsing, predictive parsers, LL(1) grammars. BOTTOM UP

PARSING: Definition of bottom up parsing, handles, handle pruning, stack implementation of shift-reduce parsing, conflicts during shift-reduce parsing, LR grammars, LR parsers-simple LR, canonical LR(CLR) and Look Ahead LR (LALR) parsers. SYNTAX DIRECTED TRANSLATION: Syntax directed definition,

construction of syntax trees, S attributed and L-attributed definitions, translation schemes, emitting a translation. INTERMEDIATE CODE GENERATION: intermediate forms of source programs– abstract syntax tree, polish notation and three address code, types of three address statements and its

implementation, syntax directed translation into three- address code, translation of simple statements, Boolean expressions and flow-of-control statements. CODE OPTIMIZATION: Organization of code optimizer, basic blocks and flow graphs, optimization of basic blocks, the principal sources of optimization,

the directed acyclic graph (DAG) representation of basic block, global data flow analysis. CODE GENERATION: Machine dependent code generation, object code forms, the target machine, a simple code generator, register allocation and assignment, peephole optimization.

Text Books :1. Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Compilers: Principles, Techniques and Tools, Second Edition, Pearson Education, 2009. 2. Allen I. Holub, Compiler Design in C, Prentice-Hall Software Series, 1993

Reference Books :1. Keith D Cooper and Linda Torczon, Engineering a Compiler, Morgan Kaufmann Publishers Elsevier Science, 2004 2. Steven S. Muchnick, Advanced Compiler Design and Implementation, Morgan Kaufmann Publishers - Elsevier Science, India, Indian Reprint 2003

Web Links :<https://nptel.ac.in/courses/106104123-Compiler Design, IIT Kanpur-Prof. Sanjeev K Aggarwal>

Course Rationale :Compiler Design is a course that introduces students to some basics in the design and implementation of compilers. This course covers the theory behind various components of a compiler such as Lexical Analyzer, Syntax and Semantic Analyzer, Intermediate code generator, Code optimization and Code generation and also the programming techniques involved to put the theory into practice, and the interfaces used to modularize the compiler.

Course Objectives :The objective of the Compiler Design course is to help students understand how compilers work, covering phases like analysis, optimization, and code generation. Students will learn basic tools and techniques to design simple compilers and apply these in practice.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Design lexical analyzers for corresponding regular expressions	PO1,PO2	3
CO2	Design lexical analyzers for corresponding regular expressions	PO2,PO3	3
CO3	Design intermediate code generation	PO1,PO3	3
CO4	Apply code optimization techniques and apply them to generate efficient code.	PO1,PO2,PO3	4
CO5	Design a simple compiler using Lex and YACC	PO3,PO5	3

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-2	COI-3	COI-4
CO1	3	Btl-2 Understanding the introduction for compilers and its phases, Understanding Finite Automata, Regular Expressions and its conversions.	Btl-3 Apply the concepts of pass and phases of translation, bootstrapping and LEX-lexical analyzer generator.	
CO2	3	Btl-2 Understand the concepts of parsing, role of parser, context free grammar, derivation trees, parse trees, ambiguity, elimination of left recursion, left factoring	Btl-3 Applying the concepts of removing ambiguity from dangling else grammar, Top down parsing , Applying the concepts of Bottom up parsing.	
CO3	3	Btl-2 Understanding the concepts of SYNTAX DIRECTED TRANSLATION	Btl-3 Applying the concepts of INTERMEDIATE CODE GENERATION and types of three address statements. Applying the concepts of syntax directed translation into three address code and flow-of-control statements.	
CO4	4	Btl-2 Understand the Concepts of basic blocks and flow graphs, optimization of basic blocks.	Btl-3 Applying the Concepts of the principal sources of optimization and DAG, Applying the Concepts of CODE GENERATION	Btl-4 Analyze lexical, syntactic, and semantic structures within compilers to evaluate code compliance, parsing techniques, and error detection.
CO5	3		Btl-3 Design the solution for real world problems using c	

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
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Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-2	INTRODUCTION TO COMPILERS: Definition of compiler, interpreter and its differences,	[1]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
2	CO1	COI-3	the phases of compiler, role of lexical analyzer,	2	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
3	CO1	COI-3	regular expressions, finite automata	[83]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
4	CO1	COI-3	from regular expressions to finite automata, pass and phases of translation	[101]	Chalk,PPT,Talk	ALM,End Semester Exam,SEM-EXAM1
5	CO1	COI-3	bootstrapping, LEX-lexical analyzer generator.	[120]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
6	CO2	COI-2	PARSING: Parsing, role of parser, context free grammar, derivations,	[159]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
7	CO2	COI-3	parse trees, ambiguity, elimination of left recursion	[171]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
8	CO2	COI-3	left factoring, eliminating ambiguity from dangling-else grammar	[179]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
9	CO2	COI-3	classes of parsing, top down parsing - backtracking, recursive descent parsing	[184]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM1
10	CO2	COI-3	predictive parsers, LL(1) grammars. BOTTOM UP PARSING: Definition of bottom up parsing	[190]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
11	CO2	COI-3	handles, handle pruning, stack implementation of	[194]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			shift-reduce parsing, conflicts during shift-reduce parsing			
12	CO2	COI-3	LR grammars, LR parsers-simple LR	[202]	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM1
13	CO2	COI-3	canonical LR(CLR) and Look Ahead LR (LALR) parsers	[210]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM1
14	CO3	COI-2	SYNTAX DIRECTED TRANSLATION: Syntax directed definition, construction of syntax trees, S attributed and L-attributed definitions,	[279]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
15	CO3	COI-3	translation schemes, emitting a translation. INTERMEDIATE CODE GENERATION: intermediate forms of source programs– abstract syntax tree	[244]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2
16	CO3	COI-3	polish notation and three address code, types of three address statements and its implementation,	[367]	Chalk,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
17	CO3	COI-3	syntax directed translation into three-address code	[372]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
18	CO3	COI-3	translation of simple statements	[387]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
19	CO3	COI-3	Boolean expressions and flow-of-control statements.	[423]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
20	CO4	COI-2	CODE OPTIMIZATION: Organization of code optimizer, basic blocks and flow graphs,	[602]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
21	CO4	COI-4	The principal sources of optimization, the directed acyclic graph (DAG)	[643]	Chalk,PPT,Talk	ALM,End Semester Exam,Home Assignment,SEM-EXAM2

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
			representation of basic block			
22	CO4	COI-4	global data flow analysis. CODE GENERATION: Machine dependent code generation, object code forms	[612]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
23	CO4	COI-3	the target machine, a simple code generator	[687]	Chalk,PPT,Talk	ALM,End Semester Exam,SEM-EXAM2
24	CO4	COI-4	register allocation and assignment	[689]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2
25	CO4	COI-4	global data analysis	[693]	Chalk,LTC,PPT,Talk	End Semester Exam,SEM-EXAM2
26	CO4	COI-4	implementation of Directed Acyclic Graph(DAG) and peephole optimizatio	[706]	Chalk,PPT,Talk	End Semester Exam,SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 INTRODUCTION TO COMPILERS: Definition of compiler, interpreter and its differences,

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	INTRODUCTION TO COMPILERS: Definition of compiler, interpreter and its differences,	2	PPT	--- NOT APPLICABLE ---
45	interpreter and its differences,	2	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 the phases of compiler, role of lexical analyzer,

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	the phases of compiler,	3	Talk	One minute paper
45	role of lexical analyzer,	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 3**Session Outcome: 1** regular expressions, finite automata

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	regular expressions	3	PPT	--- NOT APPLICABLE ---
45	finite automata	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 4**Session Outcome: 1** from regular expressions to finite automata, pass and phases of translation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	from regular expressions to finite automata	3	Talk	Brain storming session
45	pass and phases of translation	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 5**Session Outcome: 1** bootstrapping, LEX-lexical analyzer generator.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	bootstrapping	3	PPT	--- NOT APPLICABLE ---
45	LEX-lexical analyzer generator.	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 6**Session Outcome: 1** PARSING: Parsing, role of parser, context free grammar, derivations,

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	PARSING: Parsing, role of parser,	2	PPT	--- NOT APPLICABLE ---
45	PARSING: Parsing, role of parser, context free grammar, derivations,	2	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 7**Session Outcome: 1** parse trees, ambiguity, elimination of left recursion

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	parse trees, ambiguity	3	Talk	Mind Maps
45	elimination of left recursion	3	PPT	--- NOT APPLICABLE ---

5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---
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SESSION NUMBER : 8**Session Outcome:** 1 left factoring, eliminating ambiguity from dangling-else grammar

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	left factoring,	3	PPT	--- NOT APPLICABLE ---
45	eliminating ambiguity from dangling-else grammar	3	PPT	--- NOT APPLICABLE ---
45	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 9**Session Outcome:** 1 classes of parsing, top down parsing - backtracking, recursive descent parsing

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	classes of parsing, top down parsing -	3	PPT	Group Discussion
45	backtracking, recursive descent parsing	3	PPT	Group Discussion
5	conclusion, summary	3	PPT	Group Discussion

SESSION NUMBER : 10**Session Outcome:** 1 predictive parsers, LL(1) grammars. BOTTOM UP PARSING: Definition of bottom up parsing

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	predictive parsers,	3	PPT	--- NOT APPLICABLE ---

45	LL(1) grammars. BOTTOM UP PARSING: Definition of bottom up parsing	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 handles, handle pruning, stack implementation of shift-reduce parsing, conflicts during shift-reduce parsing

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	handles, handle pruning,	3	PPT	--- NOT APPLICABLE ---
45	stack implementation of shift-reduce parsing, conflicts during shift-reduce parsing	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 LR grammars, LR parsers-simple LR

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	LR grammars, LR parsers	3	PPT	--- NOT APPLICABLE ---
45	simple LR	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 1 canonical LR(CLR) and Look Ahead LR (LALR) parsers

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	canonical LR(CLR)	3	PPT	--- NOT APPLICABLE ---
45	Look Ahead LR (LALR) parsers	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 1 SYNTAX DIRECTED TRANSLATION: Syntax directed definition, construction of syntax trees, S attributed and L-attributed definitions,

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	SYNTAX DIRECTED TRANSLATION: Syntax directed definition, construction of syntax trees, S attributed	2	PPT	--- NOT APPLICABLE ---
45	S attributed and L-attributed definitions,	2	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 1 translation schemes, emitting a translation. **INTERMEDIATE CODE GENERATION:** intermediate forms of source programs– abstract syntax tree

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	translation schemes, emitting a translation.	3	PPT	Quiz/Test Questions
45	INTERMEDIATE CODE GENERATION: intermediate forms of source programs– abstract syntax tree	3	PPT	Quiz/Test Questions
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 1 polish notation and three address code, types of three address statements and its implementation,

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	polish notation and three address code, types of three address statements and its implementation,	3	PPT	Debate
45	implementation,	3	PPT	Debate
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 1 syntax directed translation into three- address code

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	syntax directed translation into three- address code	3	PPT	--- NOT APPLICABLE ---
45	syntax directed translation into three- address code	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 1 translation of simple statements

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	translation of simple statements	3	PPT	--- NOT APPLICABLE ---
45	translation of simple statements	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE

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SESSION NUMBER : 19

Session Outcome: 1 Boolean expressions and flow-of-control statements.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	Boolean expressions	3	PPT	--- NOT APPLICABLE ---
45	flow-of-control statements.	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 1 CODE OPTIMIZATION: Organization of code optimizer, basic blocks and flow graphs,

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	CODE OPTIMIZATION: Organization of code optimizer	2	PPT	--- NOT APPLICABLE ---
45	basic blocks and flow graphs,	2	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 1 The principal sources of optimization, the directed acyclic graph (DAG) representation of basic block

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	The principal sources of optimization	4	PPT	Seminars

45	the directed acyclic graph (DAG) representation of basic block	4	PPT	Seminars
5	conclusion, summary	4	PPT	Seminars

SESSION NUMBER : 22

Session Outcome: 1 global data flow analysis. CODE GENERATION: Machine dependent code generation, object code forms

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	global data flow analysis.	4	PPT	--- NOT APPLICABLE ---
45	CODE GENERATION: Machine dependent code generation, object code forms	4	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	4	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 23

Session Outcome: 1 the target machine, a simple code generator

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	the target machine, a simple code generatorthe target machine,	3	PPT	Leading question
45	a simple code generator	3	PPT	Leading question
5	conclusion, summary	3	PPT	Leading question

SESSION NUMBER : 24

Session Outcome: 1 register allocation and assignment

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---

45	register allocation and assignment	4	PPT	--- NOT APPLICABLE ---
45	register allocation and assignment	4	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	4	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 25**Session Outcome: 1** global data analysis

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	global data analysis	4	PPT	--- NOT APPLICABLE ---
45	global data analysis	4	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	4	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 26**Session Outcome: 1** implementation of Directed Acyclic Graph(DAG) and peephole optimizatio

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Attendance	1	Talk	--- NOT APPLICABLE ---
45	implementation of Directed Acyclic Graph(DAG) and peephole optimizatio	4	PPT	--- NOT APPLICABLE ---
45	impleGraph(DAG) and peephole optimizatio	3	PPT	--- NOT APPLICABLE ---
5	conclusion, summary	3	PPT	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan: NO Delivery Plan Exists**Tutorial Session wise Teaching – Learning Plan**

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Elimination of white spaces and comments	CO5
2	Building Lexical Analyzer	CO5
3	Lexical Analyzer using the LEX tool	CO5
4	Elimination of Left Recursion from given CFG	CO5
5	Program to compute FIRST set	CO5
6	Program to compute FOLLOW set	CO5
7	Construction Of LL(1) Parsing table	CO5
8	Build Shift-Reduce parser	CO5
9	Implementation of LR parse	CO5
10	Quadruple notation	CO5
11	Implement code optimization technique	CO5
12	Implement a module for generating intermediate code ..	CO5
13	Implement basic error detection and reporting system	CO5

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Elimination of white spaces and comments

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Elimination of white spaces and comments	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2**Session Outcome: 2** Building Lexical Analyzer

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Building Lexical Analyzer	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3**Session Outcome: 1** Lexical Analyzer using the LEX tool

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Lexical Analyzer using the LEX tool	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4**Session Outcome: 1** Elimination of Left Recursion from given CFG

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Elimination of Left Recursion from given CFG	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5**Session Outcome: 1** Program to compute FIRST set

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Program to compute FIRST set	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6**Session Outcome: 1** Program to compute FOLLOW set

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Program to compute FOLLOW set	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7**Session Outcome: 1** Construction Of LL(1) Parsing table

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Construction Of LL(1) Parsing table	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8**Session Outcome: 1** Build Shift-Reduce parser

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
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10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Build Shift-Reduce parser	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9**Session Outcome: 1** Implementation of LR parse

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Implementation of LR parse	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10**Session Outcome: 1** Quadruple notation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Quadruple notation	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11**Session Outcome: 1** Implement code optimization technique

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---

60	Implement code optimization technique	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12**Session Outcome: 1** Implement a module for generating intermediate code ..

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Implement a module for generating intermediate code ..	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 13**Session Outcome: 1** Implement basic error detection and reporting system

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
10	Attendance	1	Talk	--- NOT APPLICABLE ---
60	Implement basic error detection and reporting system	3	Talk	--- NOT APPLICABLE ---
30	viva voice	3	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan: NO Delivery Plan Exists**Skilling Session wise Teaching – Learning Plan**

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
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COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Tutorial	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Lab	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Skilling	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
Tue	Theory	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Tutorial	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Lab	-	-	--	--	--	--	-	-	-
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	Skilling	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
Wed	Theory	-	-	--	--	--	--	-	-	-
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	Tutorial	-	-	--	--	--	--	-	-	-
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	Lab	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Skilling	-	-	--	--	--	--	-	-	-
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Thu	Theory	-	-	--	--	--	--	-	-	-
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	Tutorial	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Lab	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Skilling	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
Fri	Theory	-	-	V-S101,V-S102,V-S104,V-S204	V-S101,V-S102,V-S104,V-S204	V-S103,V-S201,V-S202,V-S203	V-S103,V-S201,V-S202,V-S203	-	-	-
		-	-					-	-	-
	Tutorial	-	-	--	--	--	--	-	-	-
		-	-					-	-	-
	Lab	-	-	V-S103,V-S103,V-S201,V-S201,V-S201,V-S202,V-S202,V-S203,V-S203,V-S203	V-S103,V-S103,V-S201,V-S201,V-S201,V-S202,V-S202,V-S203,V-S203,V-S203	V-S101,V-S101,V-S102,V-S102,V-S102,V-S104,V-S104,V-S104,V-S204,V-S204,V-S204	V-S101,V-S101,V-S102,V-S102,V-S102,V-S104,V-S104,V-S104,V-S204,V-S204,V-S204	-	-	-
		-	-					-	-	-
	Skilling	-	-	--	--	--	--	-	-	-
		-	-					-	-	-

Sat	Theory	-	-	--	--	--	--	-	-	-
	Tutorial	-	-	--	--	--	--	-	-	-
	Lab	-	-	--	--	--	--	-	-	-
	Skilling	-	-	--	--	--	--	-	-	-
Sun	Theory	-	-	--	--	--	--	-	-	-
	Tutorial	-	-	--	--	--	--	-	-	-
	Lab	-	-	--	--	--	--	-	-	-
	Skilling	-	-	--	--	--	--	-	-	-

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4	CO5
End Semester Summative Evaluation Total= 40 %	Lab End Semester Exam	Weightage	16		90					16
		Max Marks	100							100
	End Semester Exam	Weightage	24		180	6	6	6	6	
		Max Marks	100			25	25	25	25	
In Semester Formative Evaluation Total= 24 %	Continuous Evaluation - Lab Exercise	Weightage	10		90					10
		Max Marks	25							25
	Home Assignment and	Weightage	6		30	1.5	1.5	1.5	1.5	

	Textbook	Max Marks	50			12.5	12.5	12.5	12.5	
	ALM	Weightage	8		30	2	2	2	2	
		Max Marks	50			12.5	12.5	12.5	12.5	
In Semester Summative Evaluation Total= 36 %	Lab In Semester Exam	Weightage	8		90					8
		Max Marks	50							50
	Semester in Exam-II	Weightage	14		90			7	7	
		Max Marks	50					25	25	
	Semester in Exam-I	Weightage	14		90	7	7			
		Max Marks	50			25	25			

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course

In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Radhika Chintala	L	104-MA	-	-	-	-
Radhika Chintala	P	104-A	-	-	-	-
Nagamalleswari Dubba	L	102-MA	-	-	-	-
Nagamalleswari Dubba	P	102-A	-	-	-	-
KRISHNA CHAITANYA GOGINENI	P	101-C	-	-	-	-
Anjali Devi Swarna	L	103-MA	-	-	-	-
Anjali Devi Swarna	P	103-A	-	-	-	-

Maganti Syamala	P	201-C	-	-	-	-
Tirandasu Kumar	P	103-B	-	-	-	-
Asesh Tripathy	L	201-MA	-	-	-	-
Asesh Tripathy	P	201-A	-	-	-	-
Vuyyuru Reddy	P	103-C	-	-	-	-
Dayanidhi Mohapatra	P	204-C	-	-	-	-
KUCHIPUDI PRASANTH KUMAR	L	202-MA	-	-	-	-
KUCHIPUDI PRASANTH KUMAR	P	202-A	-	-	-	-
SETTI RAJESWARI	P	202-C	-	-	-	-
KAVITHA SARIHADDU	L	101-MA	-	-	-	-
KAVITHA SARIHADDU	P	101-A	-	-	-	-
ANUMULA SRUTHI	P	202-B	-	-	-	-
KISHORE DASARI	P	201-B	-	-	-	-
surya chebrolu	P	203-B	-	-	-	-
PINAPATI RAO	P	203-C	-	-	-	-
BADUGU SAMATHA	P	102-B	-	-	-	-
D Pramodh Krishna	L	203-MA	-	-	-	-
D Pramodh Krishna	P	203-A	-	-	-	-
Madabattula Kumar	P	101-B	-	-	-	-
Badde Prakash	P	104-B	-	-	-	-
BILLA RAO	P	102-C	-	-	-	-
AMIT KAR	P	204-B	-	-	-	-
SATEESH G	L	204-MA	-	-	-	-
SATEESH G	P	204-A	-	-	-	-
Geetha M	P	104-C	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(KAVITHA SARIHADDU)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of CSE-1

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal) [object HTMLDivElement]